

- 3 (a) $V = h \times A$
 $m = V \times \rho$ B1
 $W = h \times A \times \rho \times g$ B1
 $P = F / A$ B1
 $P = h\rho g$
 P is proportional to h if ρ is constant (and g) B1 [4]
- (b) density changes with height B1
hence density is not constant with link to formula B1 [2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2012	9702	23

1 (a) electric field strength is the force per unit positive charge (acting on a stationary